

A Multi-Modal Quantification of Parkinson's Medication Response Using Kinematic Rigidity Profiling and Acoustic Feature Regression: A Proof-of-Concept Study

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Abstract—PD affects over 60% of people above 50, and its UPDRS-based monitoring demands clinical visits and lengthy evaluations, making monitoring costly and inconvenient. This paper presents a first integrated multi-modal telemonitoring system for Parkinson's disease, combining kinematic rigidity profiling with speech-based regression modelling to estimate medication response severity in Parkinson. Wrist rigidity is quantified using synchronized Inertial Measurement Units (IMUs), while acoustic features extracted from sustained phonation are used to predict UPDRS scores via an XGBoost regression model. Both modalities are fused to generate a unified Medication Response Score (MRS), providing a continuous estimation of patient state rather than categorical ON/OFF detection. This multi-modal fusion will be performed using an application-based system responsible for all data processing and user interaction. For the speech part, we used the largest known PD speech database, comprising nearly 6,000 recordings from 42 patients, gathered over a six-month, multi-centre trial. Preliminary prototyping, hardware validation and algorithm of our regression model achieved R^2 0.887 and MSE 11.49, demonstrating strong alignment with clinician-rated UPDRS. A Bland–Altman analysis confirmed minimal bias acceptable agreement limits. These results highlight the feasibility of a non-invasive, low-cost, portable system for remote UPDRS tracking in Parkinson's telemonitoring. Future work will integrate full sensor–speech processing on-device with clinical-grade validation.

Index Terms—Parkinson's Disease (PD), Inertial Measurement Unit (IMU), Telemonitoring, Rigidity, Speech Analysis, Machine Learning, Multi-Modal Fusion.

I. INTRODUCTION

Parkinson's disease (PD) is a neurodegenerative condition that poses major challenges for long-term monitoring [1], [2]. It is a neurological condition that impairs motor function, resulting in tremors, stiffness, and slow motion.

Conventional assessments such as the UPDRS are resource-demanding, making regular monitoring impractical [3].

A crucial component of disease treatment, especially as Parkinson's illness progresses, is the “on–off” behavior as highlighted by Marsden and Parkes in [4]. This refers to the variability in a patient's reaction to dopaminergic therapy, including levodopa, throughout the day. The “On” State is the period when medication is effective, and symptoms are well-controlled. Patients experience improved mobility, reduced rigidity, and less tremor. They can perform daily activities with relative ease. The transition to the “on” state typically occurs 30–45 minutes after taking medication and can last for a few hours. The “Off State” is the period when medication effects have worn off, and motor symptoms reappear. Patients may experience increased tremor, bradykinesia (slowness of movement), and severe rigidity, leading to difficulties with walking and performing tasks. Motor disability in Parkinson's disease tends to worsen over time, as shown in a population based study [5], [6]. The “off” state is also often associated with non-motor symptoms, such as anxiety, restlessness, and mood changes leading to misdiagnosis at times.

In Parkinson's disease, medication—particularly levodopa—improves rigidity and speech, as both issues arise from dopamine deficiency in the brain. The medicine restores dopamine, aiding in the restoration of motor control for the muscles involved in limb movement and speech articulation improving overall movement. Current Parkinson's disease biomedical technologies frequently address isolated symptoms such as tremors or gait [19], constraining comprehension of the disease as a whole. For example, a tremor-only device cannot monitor speech decline, hence overlooking symptom correlations such as stiffness impacting speech and diminishing the efficacy of adaptive therapy.

Our proposed model combines speech and rigidity assessment, capturing variations in motor and verbal difficulties

across on/off states. Our objective is to overcome this by creating an integrated device that combines rigidity and speech monitoring. We will leverage the direct link between increased muscular rigidity and impaired speech to track both simultaneously. Our device will also provide a comprehensive view of a patient's "on-off" states, helping clinicians make better treatment decisions. By using this combined data, the device can deliver targeted, real-time interventions, such as vibrotactile stimulation for rigidity or auditory cues for speech.

II. RELATED WORK

Wearable devices are a key research area for objectively measuring rigidity in Parkinson's disease, which is a symptom that doctors usually check for subjectively. These methods are designed to measure how resistant a patient is to passive movement, which is a key sign of the disease. Although our main focus is on recent developments (2020–2025), it is important to acknowledge the earlier wearable-based research from 2015–2019, which laid the groundwork for today's digital biomarker and telemonitoring approaches in Parkinson's disease. A series of influential contributions during this period collectively demonstrated both the feasibility and the challenges of wearable-based PD monitoring. In Parkinson studies, many rigidity studies [7] employ accelerometers and gyroscopes. Researchers can assess the smoothness, range of motion, and velocity during pronation/supination tasks or finger tapping by inserting these sensors on the wrist, fingers, or limbs. Nikolic' et al. demonstrated the feasibility of IMU-based movement tracking in Parkinson's patients [8]. Researchers have created methods to monitor limb movement force or torque. These devices use micro motors and transducers to measure a patient's limb's passive flexion and extension resistance [9]. Recent advances show that IoT devices like smartphones, smartwatches, and wearable inertial sensors now enable continuous, real-world monitoring of Parkinson's symptoms at home. TSilva de Lima et al. [26] showed that smartphones and smartwatches can be used at scale for Parkinson's monitoring, with nearly 1000 participants providing accelerometer and activity data in the Parkinson at home study. Their findings proved that everyday consumer devices are capable of supporting long-term, real-world symptom tracking beyond clinic visits, although challenges like inconsistent data quality and lack of standardization still remained. [10].

Speech analysis is a non-invasive and increasing method for monitoring PD since most patients develop hypokinetic dysarthria. In [11] advanced speech algorithms have been proposed for accurate PD detection. Algorithms are used to analyze the voice recordings to extract features such as fundamental frequency f_0 , jitter, and shimmer, which quantify pitch and amplitude fluctuation, respectively. Researchers use speech databases from Parkinson's patients to train ML models—like SVMs [12] and neural networks—for disease classification and severity assessment. These systems frequently employ activities such as prolonged vowel phonation or rapid syllable repetition.

Therefore, our objective is to modify and work on current devices. Our aim is to address this by developing a unified device that integrates stiffness and voice tracking assisting physicians in making improved treatment decisions.

III. METHODOLOGY

A. Rigidity Path Simulation

Our proposed methodology outlines the design and implementation of a low-cost, portable device for the objective estimation of wrist rigidity in individuals with Parkinson's disease in "off" and "on" medication states. The primary inspiration for our device's objective rigidity measurement comes from the PD meter [13], a force-based mechatronic system built for this purpose. Our system measures stiffness and damping, which are currently assessed subjectively in clinical settings. Our system will use hardware and software to gather and analyze wrist motion data for at-home telemonitoring.

Our approach introduces a validated, sensor-based system to objectively assess wrist stiffness in Parkinson's disease (PD) patients for "On-Off" medication states. This system is designed for home-based telemonitoring, directly addresses the limitations of subjective clinical assessments by providing a repeatable and quantitative measure of dynamic stiffness.

The core of our approach is the use of two synchronized high-quality Inertial Measurement Units (IMUs), a pre-programmed STM32 microcontroller, and a data-processing framework. Wearable IMUs consist of inertial sensors utilizing Micro-Electro-Mechanical Systems (MEMS) technology [14].

The system's foundation aims to be cost-effective, portable device including two synchronized high-quality Inertial Measurement Units (IMUs), an integrated STM32 microcontroller, and a data storage unit.

The stiffness estimation will follow to a three-step procedure: data collecting, on-board processing, and parameter estimation. The rigidity assessment procedure has been executed and corroborated via a proof-of-concept prototype. The apparatus comprises two ICM-42688-P IMUs that are firmly affixed to the patient's forearm and hand. We have achieved effective communication and synchronization between the two sensors and a microcontroller in our present study.

A caregiver performs a standardized, passive wrist flexion-extension technique. Throughout this procedure, the IMUs transmit three-dimensional angular acceleration data (α) to the microcontroller.

Biomechanically, Rigidity is the resistance to passive movement, can be quantified using rotational dynamics.as:

$$\tau = I \cdot \alpha, \quad (1)$$

where τ represents torque, I is the hand's moment of inertia, and α denotes the angular acceleration measured by the IMU.

The on-board STM32 microcontroller processes the synchronized data streams, filtering out noise and computing the relative angular motion between the hand and forearm. The critical step in our methodology is the calculation of the patient's total body mass m_{total} and hand length l_{hand} , that

are used to estimate the hand's mass and, subsequently, its moment of inertia (I).

Here, the hand mass is approximated using anthropometric ratios:

$$m_{\text{hand}} = r_{\text{hand}} \cdot m_{\text{total}}, \quad (2)$$

where r_{hand} is the average percentage of hand mass relative to body mass (0.5–0.6%), and m_{total} is the total body weight.

Combined with segment length, the moment of inertia is computed as:

$$I = m_{\text{hand}} \cdot d^2, \quad (3)$$

where d denotes the distance from the wrist joint to the hand's center of mass.

Finally, rigidity index (RI) is obtained as the ratio of torque to angular displacement (θ):

$$RI = \frac{\tau}{\theta} \quad (4)$$

Ultimately, the complete calculation procedure is carried out automatically by the software running on the microcontroller and the backend of our web application, with the patient's weight serving as a user-provided input.

Fig. 1 illustrates the internal working of our rigidity analysis device. The STM32 microcontroller acts as the central processor, handling synchronized data acquisition, filtering, and computation of relative angular motion from the IMU sensors. The system subsequently calculates the Rigidity Index (RI) utilizing the calculated torque and angular displacement, then securely archives all timestamped data onto a MicroSD card and facilitates wireless data transmission.

We utilized the following components to build our proof-of-concept prototype as shown in Table I. Our connectivity and data acquisition tests were successful.

TABLE I
HARDWARE COMPONENTS

Component	Functionality and Status
STM32 Microcontroller	Main processing unit, programmed to manage IMU data acquisition and preliminary calculations.
2x ICM-42688-P IMUs	High-precision sensors with reliable communication and synchronization with the STM32.
MicroSD Card Module	On-board memory used to store raw sensor data, verified for proper data writing.
Rechargeable Li-Po Battery	Provides portable power, enabling wearable, cable-free design.
Bluetooth Module	Wireless connectivity and data transfer from the microcontroller.

Fig. 2 provides a clear representation of the proposed hardware design for our rigidity assessment system. The diagram illustrates the placement of two IMU sensors—one on the hand and the other on the forearm—while the microcontroller and SD card module are situated on the forearm. IMUs are placed on the hand and forearm to accurately measure the relative motion between the two segments. This placement

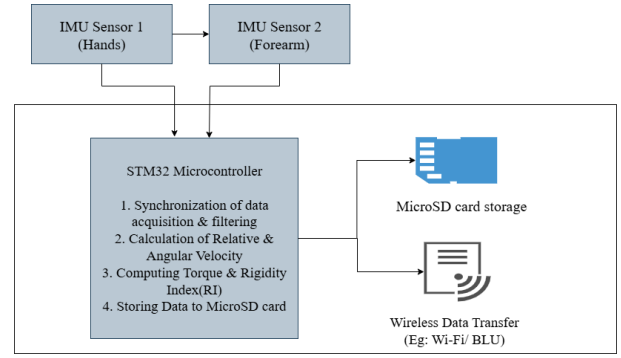


Fig. 1. Framework of the wearable control model



Fig. 2. Conceptual Wearable Prototype for Rigidity Assessment.

is crucial because wrist rigidity is defined as the resistance to passive flexion and extension, which occurs at the wrist joint. This configuration is specifically engineered to document the relative motion between the two pieces, which is essential for our stiffness calculations. Our proposed methodology for evaluating wrist rigidity is predicated on the principle of a quantitative, at-home evaluation.

B. Speech Path Analysis

Parkinson's disease (PD) often causes dysphonia, a vocal impairment caused by neurological control deficits in the laryngeal and respiratory muscles [15]. Acoustic metrics from speech signals can quantify impairment by detecting variations in vocal fold vibration, pitch, and amplitude [17]. This research proposes that digital voice signal processing can create a model to predict a patient's state. The study aimed to create a statistical relationship between dysphonia measurements and the UPDRS, providing a non-invasive, objective assessment of illness severity.

1) *Dataset Utilization* : The paper [16] investigated the feasibility of a non-invasive telemonitoring system utilizing voice analysis to monitor disease progression. This foundational work, which supplies the dataset for the current analysis, proposes that minor alterations in vocal function can reliably indicate greater brain changes linked to Parkinson's disease.

TABLE II
CORRELATION OF DYSPHONIA METRICS WITH MOTOR AND TOTAL
UPDRS SCORES

Metric	Motor Corr.	Total Corr.	Description
MDVP: Jitter (%)	0.124	0.125	Jitter as a percentage (KP-MDVP)
MDVP: Jitter (Abs)	0.072	0.103	Absolute jitter in μs (KP-MDVP)
MDVP: RAP	0.105	0.107	Relative Amplitude Perturbation
MDVP: PPQ	0.120	0.117	Five-cycle Period Perturbation Quotient
Jitter: DDP	0.105	0.107	Cycle-to-cycle difference normalized by mean period
MDVP: Shimmer	0.138	0.139	Local shimmer (KP-MDVP)
MDVP: Shimmer (dB)	0.139	0.139	Local shimmer in decibels
Shimmer: APQ3	0.116	0.122	Amplitude Perturbation Quotient (3 cycles)
Shimmer: APQ5	0.123	0.127	Amplitude Perturbation Quotient (5 cycles)
MDVP: APQ	0.166	0.163	Amplitude Perturbation Quotient (11 cycles)
Shimmer: DDA	0.116	0.122	Difference of sequential amplitude variations
NHR	0.131	0.139	Noise-to-Harmonics Ratio
HNR	-0.159	-0.163	Harmonics-to-Noise Ratio
RPDE	0.112	0.143	Recurrence Period Density Entropy
DFA	-0.131	-0.141	Detrended Fluctuation Analysis
PPE	0.160	0.152	Pitch Period Entropy

All measurements are obtained from sustained vowel phonations. Values show Spearman correlation (p-value < 0.0001) with motor and total UPDRS scores. MDVP refers to the Kay Pentax Multidimensional Voice Program.

This work utilizes the publicly available dataset initially identified by Tsanas et al. [16] including 42 participants with Parkinson’s disease (PD) monitored via telemedicine over six months. Using the Intel At-Home Testing Device (AHTD), 5,923 sustained “ahh. . .” phonations were recorded. Each phonation was analyzed to extract 16 dysphonia-related acoustic features, covering classical linear measures (e.g., jitter, shimmer, harmonics-to-noise ratio) and newer nonlinear measures (e.g., Recurrence Period Density Entropy, Detrended Fluctuation Analysis) forming a $5,923 \times 16$ matrix, serves as the input for our predictive model, consistent with the foundational work [16]. These qualities were developed in order to show how PD affects the voice, and they were first linked to the Unified Parkinson’s illness Rating Scale (UPDRS) so that illness progression could be tracked.

In our study, Table II summarizes key voice-based metrics for Parkinson’s disease assessment, that provides a quantitative foundation for evaluating speech impairments. In this work, we adapt this dataset for a different objective. In this work, we no longer transform UPDRS into binary ON/OFF medication states — instead we predict continuous Total UPDRS scores using regression models. However, The core methodology for data acquisition and feature extraction aligns with the original paper to maintain comparability.

2) *Data Pre-processing and Model Development:* All speech features were scaled to have zero mean and unit variance for comparability. Missing values were checked, and recordings with artifacts were excluded as per the original protocol. The original paper examined various linear and nonlinear regression techniques, including Least Squares (LS), Classification and Regression Tree (CART) and Iteratively Re-weighted Least Squares (IRLS). We followed their cross-validation methodology by partitioning the data into a training set (5,331 phonations) and a testing set (592 phonations), with our principal alteration being the use of a machine learning algorithm. The dataset was modelled with a continuous Total UPDRS score (Y) using a reduced, symptom-specific feature

subset (X) including: Age, Sex, Motor-UPDRS, Jitter%, Shimmer(dB), NHR, HNR, RPDE, DFA, and PPE. This feature group was chosen because each parameter captures vocal degradation linked to Parkinson’s pathology—specifically reduced phonatory stability, irregular vocal fold vibration, and neuromotor dysfunction that manifest in dysphonic speech.

3) *Model Selection and Performance Assessment:* We conducted a comprehensive comparison investigation of multiple machine learning models, including XGBoost, CatBoost, and HistGB, to predict UPDRS scores based on dysphonia characteristics. Among them, the XGBoost delivered the best performance with: $R^2 = 0.8875$ (88.7% variance explained), $MSE = 11.49$, Train $R^2 = 0.8860$ showing strong generalization and no signs of overfitting as shown in Fig 4.

To assess clinical reliability, we applied a Bland–Altman [24] comparison as shown in Fig. 5 a widely accepted benchmark for validating biomedical scoring systems. More than 95% of the predicted values lay within the ± 1.96 SD limits of agreement, demonstrating close alignment between model-derived UPDRS scores and clinician assessments, without evidence of systematic bias.

These results are consistent with the analytical framework proposed by Bland Altman and later standardized in biomedical evaluation literature [25]. This supports the reliability of our telemonitoring model for UPDRS severity estimation in both clinical and remote use. Fig 5 shows a SHAP feature-impact plot highlighting dominant predictors of UPDRS severity.

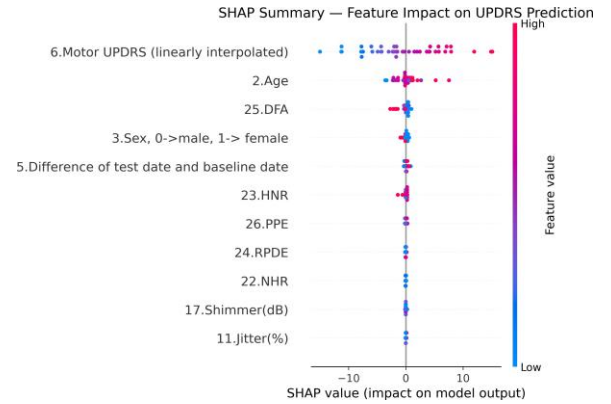


Fig. 3. SHAP summary plot showing feature impact on UPDRS prediction, with Motor-UPDRS and Age as dominant predictors and dysphonia features contributing moderately, without bias.

Overall, the system as shown in Fig 6.shows strong potential as a non-invasive, AI-enabled tool for routine Parkinson’s disease tracking research. A compact and affordable INMP441 MEMS microphone will capture the speech signal, providing clean audio suitable for extracting acoustic markers in Parkinson’s assessment. [27].

C. Multi-Modal Fusion & Integration

After SD-card upload, the backend starts multi-modal fusion, processing both IMU and audio data. Rigidity features

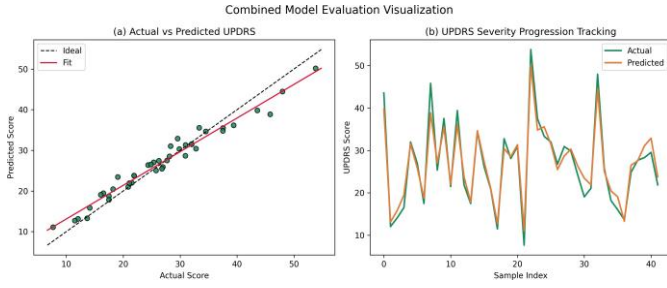


Fig. 4. Comparison of UPDRS prediction and progression tracking, demonstrating close agreement with clinical ratings.

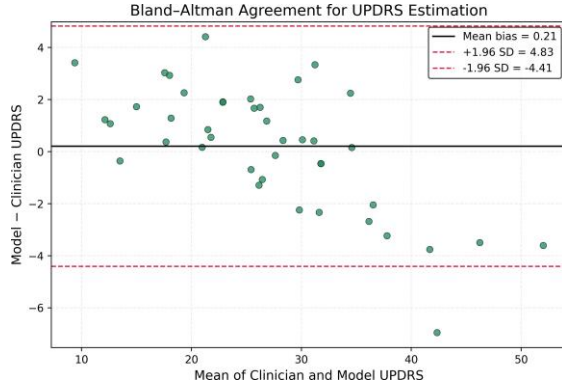


Fig. 5. Bland-Altman agreement analysis for UPDRS estimation showing negligible mean bias and 95% of samples within clinical LoA.

are computed from IMU motion using individual anthropometric parameters, and the speech recordings are processed through our validated XGBoost model to predict Total UPDRS severity. We adopt a weighted late fusion scheme to compute a single, comprehensive *Medication Response Score (MRS)*. This score is obtained as a normalized combination of the rigidity index (RI) and the speech-derived ON-probability (P_{speech}). The formulation is given by:

$$MRS = w_r \cdot RI_{\text{norm}} + w_s \cdot P_{\text{speech}}, \quad (5)$$

where RI_{norm} and P_{speech} are the normalized rigidity and speech scores, respectively. The tunable weights w_r and w_s can be adjusted by a clinician to prioritize either modality depending on the patient profile.

The following are the steps for the rehabilitation-Oriented Framework as shown in Fig 7:

a) Correlated Symptom Expression: Our methodology recognizes the frequent co-occurrence of stiffness and speech deterioration during Parkinson's "off" phases. By recording both, our equipment offers a more comprehensive and precise evaluation of a patient's overall condition.

b) Telemonitoring: The system enables patients to do tests at home, with data automatically transmitted to a web application for clinician analysis.

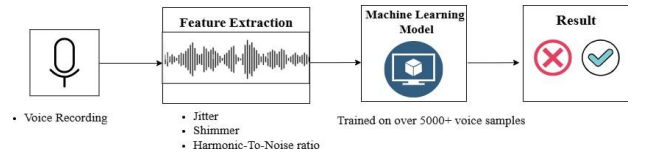


Fig. 6. Speech-Based Analysis Workflow

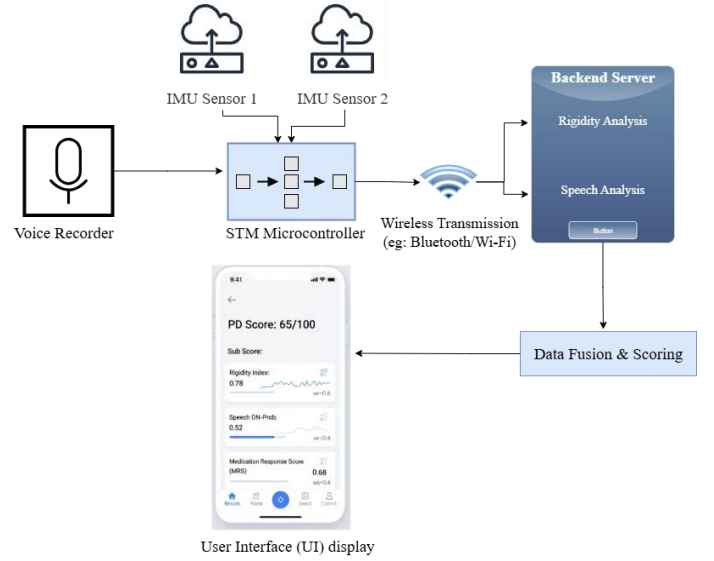


Fig. 7. Shows the flowchart of the proposed system. It shows details the data acquisition from the IMU sensors and voice recorder, their transmission to the STM32 microcontroller, and subsequent wireless transfer to a backend server. The server processes the rigidity and speech data separately before a final data fusion step. The output is a comprehensive score displayed on a user interface for clinical review.

IV. CONCLUSION & FUTURE WORK

We have successfully conducted a preliminary testing to verify the fundamental hardware and firmware of our gadget utilizing an Arduino Uno for prototyping prior to our shift to the more resilient STM32 microcontroller for on-board processing. The tests verified that the microcontroller can consistently acquire and interpret synchronized angular acceleration data from two IMU sensors, illustrating the viability of our sensor-based methodology for stiffness analysis. Additionally, we developed a regression model using the audio data, improving the system's capability to predict and quantify patient responses. Our system combines two complementary methods: a sensor-based method for measuring wrist rigidity and a machine learning model for analyzing speech. The rigidity technique offers a quantitative, objective evaluation, whereas the ML algorithm precisely predicts medication response through continuous UPDRS severity estimation. This multi-modal system combines all inputs into one Medication Response Score (MRS), which gives a complete and integrated picture of the patient's state for successful telemonitoring at home.

Our future work will focus on two key areas: developing a



Fig. 8. Experimental configuration featuring dual IMU units linked to an Arduino Uno, with live inertial data streaming to a laptop for code execution and communication validation.

fully portable device and its accompanying web application. Subsequently, developing performing benchtop validation for biomechanical accuracy, and conducting pilot clinical trials using UPDRS as the evaluation standard.

REFERENCES

- [1] F. Stocchi, D. Bravi, A. Emmi, and A. Antonini, "Parkinson disease therapy: current strategies and future research priorities," *Nature Reviews Neurology*, vol. 20, no. 12, pp. 695–707, 2024.
- [2] E. R. Dorsey, T. Sherer, M. S. Okun, and B. R. Bloem, "The emerging evidence of the Parkinson pandemic," *Journal of Parkinson's Disease*, vol. 8, no. S1, pp. S3–S8, 2018.
- [3] E. J. Hill, C. G. Mangleburg, I. Alfradique-Dunham, B. Ripperger, A. Stillwell, H. Saade, and J. M. Shulman, "Quantitative mobility measures complement the MDS-UPDRS for characterization of Parkinson's disease heterogeneity," *Parkinsonism Related Disorders*, vol. 84, pp. 105–111, 2021.
- [4] C. D. Marsden and J. D. Parkes, "On-off effects in patients with Parkinson's disease on chronic levodopa therapy," *The Lancet*, vol. 307, no. 7954, pp. 292–296, 1976.
- [5] P. Bugalho, F. Ladeira, R. Barbosa, J. P. Marto, C. Borbinha, L. da Conceição, and M. Fernandes, "Progression in Parkinson's disease: variation in motor and non-motor symptoms severity and predictors of decline in cognition, motor function, disability, and health-related quality of life as assessed by two different methods," *Movement Disorders Clinical Practice*, vol. 8, no. 6, pp. 885–895, 2021.
- [6] S. S. Paul, L. E. Dibble, G. N. Olivier, C. Walter, K. Duff, and S. Y. Schaefer, "Dopamine replacement improves motor learning of an upper extremity task in people with Parkinson disease," *Behavioural Brain Research*, vol. 377, p. 112213, 2020.
- [7] F. Caramia, E. D'Angelantonio, L. Lucangeli, and V. Camomilla, "Validation of low-cost IMUs for telerehabilitation exercises," *Sensors*, vol. 25, no. 10, p. 3129, 2025.

- [8] J. Nikolic', S. Radovanovic', and R. Nikic', "Using of inertial measurement unit (IMU) in movement analysis in persons with Parkinson's disease," *Approaches and Models in Special Education and Rehabilitation – Thematic Collection of International Importance*, pp. 361–377, 2020.
- [9] G. A. Zito, S. M. Gerber, P. Urwyler, M. J. Shamsollahi, N. Pal, D. Benninger, and T. Nef, "Development and pilot testing of a novel electromechanical device to measure wrist rigidity in Parkinson's disease," in *Proc. 40th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, Honolulu, HI, USA, Jul. 2018, pp. 4885–4888.
- [10] E. E. Stefanou, P. Bitilis, G. Bouchouras, and K. Kotis, "Collecting, integrating and processing IoT sensor data on edge devices for PD monitoring: A scoping review," *Applied Sciences*, vol. 15, no. 19, p. 10541, 2025.
- [11] M. A. A. Albadr, M. Ayob, S. Tiun, R. Z. Homod, F. T. Al-Dhief, and M. H. Mutar, "Parkinson's disease diagnosis by voice data using particle swarm optimization-extreme learning machine approach," *Multimedia Tools and Applications*, vol. 84, no. 23, pp. 26843–26876, 2025.
- [12] D. Mulhari, D. La Placa, C. Rovito, A. Celesti, and M. Villari, "Deep learning applications in telerehabilitation speech therapy scenarios," *Comput. Biol. Med.*, vol. 148, p. 105864, Feb. 2022.
- [13] L. Raiano, G. di Pino, L. di Biase, M. Tombini, N. L. Tagliamonte, and D. Formica, "PDMeter: A wrist wearable device for an at-home assessment of the Parkinson's disease rigidity," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 28, no. 6, pp. 1325–1333, Jun. 2020.
- [14] T. Noushin, M. S. Parvez, P. Biswas, M. E. H. Bhuiyan, M. A. A. Rumon, and S. Tabassum, "MEMS in twenty-first-century healthcare," in *Micro- and Nano-Systems in 21st-Century: Designs, Developments, Applications and Perspective*, Singapore: Springer Nature Singapore, 2025, pp. 71–132.
- [15] E. Schalling, "Dysarthria and dysphonia in Parkinson's disease," in *The Oxford Handbook of Communication Disorders in Neurodegenerative Diseases*, 2025, ch. 17.
- [16] A. Tsanas, M. Little, P. McSharry, and L. Ramig, "Accurate telemonitoring of Parkinson's disease progression by non-invasive speech tests," *Nature Precedings*, pp. 1–1, 2009.
- [17] L. Franz, A. Da Canal, G. Spinato, G. Baracca, E. Lucchini, C. de Filippis, and G. Marioni, "Tele-voice therapy: Application of the telemedicine paradigm to speech therapy for dysphonia—A systematic review," *American Journal of Otolaryngology*, Art. no. 104626, 2025.
- [18] L. Shahgholi, S. De Jesus, S. S. Wu, Q. Pei, A. Hassan, M. J. Armstrong, et al., "Hospitalization and rehospitalization in Parkinson disease patients: Data from the National Parkinson Foundation Centers of Excellence," *PLoS One*, vol. 12, no. 7, p. e0180425, 2017.
- [19] E. Abdulhay, N. Arunkumar, K. Narasimhan, E. Vellaippan, and V. Venkatraman, "Gait and tremor investigation using machine learning techniques for the diagnosis of Parkinson disease," *Future Generation Computer Systems*, vol. 83, pp. 366–373, 2018.
- [20] M. D. R. Ferreira-Sa'nchez, M. Moreno-Verdu', and R. Cano-de-La-Cuerda, "Quantitative measurement of rigidity in Parkinson's disease: a systematic review," *Sensors*, vol. 20, no. 3, p. 880, 2020.
- [21] E. M. Cheng, K. Swartztrauber, A. D. Siderowf, M. S. Eisa, M. Lee, S. Vassar, and B. G. Vickrey, "Association of specialist involvement and quality of care for Parkinson's disease," *Mov. Disord.*, vol. 22, no. 4, pp. 515–522, Mar. 2007.
- [22] L. J. W. Evers, J. H. Krijthe, M. J. Meinders, B. R. Bloem, and T. M. Heskes, "Measuring Parkinson's disease over time: The real-world within-subject reliability of the MDS-UPDRS," *Mov. Disord.*, vol. 34, pp. 1480–1487, 2019.
- [23] M. Sica, S. Tedesco, C. Crowe, L. Kenny, K. Moore, S. Timmons, and D. S. Komaris, "Continuous home monitoring of Parkinson's disease using inertial sensors: A systematic review," *PLoS One*, vol. 16, no. 2, p. e0246528, Feb. 2021.
- [24] J. M. Bland and D. G. Altman, "Statistical methods for assessing agreement between two methods of clinical measurement," *The Lancet*, 1986.
- [25] D. Giavarina, "Understanding Bland–Altman analysis," *Biochemia Medica*, vol. 25, no. 2, pp. 141–151, 2015.
- [26] A. Prochazka, D. J. Bennett, M. J. Stephens, S. K. Patrick, R. Sears-Duru, T. Roberts, and J. H. Jhamandas, "Measurement of rigidity in Parkinson's disease," *Movement Disorders*, vol. 12, no. 1, pp. 24–32, 1997.
- [27] P. V. Pflug, "Aspects of the use of MEMS microphones in phased array systems," in *INTER-NOISE and NOISE-CON Congress and Conference Proceedings*, vol. 255, no. 2, pp. 5093–5103, Dec. 2017.

- [28] T. Liu, M. Zhang, Z. Li, H. Dou, W. Zhang, J. Yang, and X. Mu, "Machine learning-assisted wearable sensing systems for speech recognition and interaction," *Nature Communications*, vol. 16, no. 1, p. 2363, 2025.
- [29] G. J. van der Wilt, "Alternative ways of framing Parkinson's Disease: Implications for priorities in health care and biomedical research," *Industrial & Environmental Crisis Quarterly*, vol. 9, no. 1, pp. 13–48, 1995.
- [30] M. C. Barroso, G. P. Esteves, T. P. Nunes, L. M. Silva, A. C. Faria, and P. L. Melo, "A telemedicine instrument for remote evaluation of tremor: design and initial applications in fatigue and patients with Parkinson's Disease," *Biomedical Engineering Online*, vol. 10, no. 1, p. 14, 2011.
- [31] J. G. Goldman, D. Volpe, T. D. Ellis, M. A. Hirsch, J. Johnson, J. Wood, and K. Wallock, "Delivering multidisciplinary rehabilitation care in Parkinson's disease: an international consensus statement," *Journal of Parkinson's Disease*, vol. 14, no. 1, pp. 135–166, 2024.
- [32] B. R. Chen, S. Patel, T. Buckley, R. Rednic, D. J. McClure, L. Shih, ... and P. Bonato, "A web-based system for home monitoring of patients with Parkinson's disease using wearable sensors," *IEEE Transactions on Biomedical Engineering*, vol. 58, no. 3, pp. 831–836, 2010.
- [33] S. Patel, K. Lorincz, R. Hughes, N. Huggins, J. Growdon, D. Standaert, and P. Bonato, "Monitoring motor fluctuations in patients with Parkinson's disease using wearable sensors," *IEEE Transactions on Information Technology in Biomedicine*, vol. 13, no. 6, pp. 864–873, 2009.